

DAYANANDA SAGAR COLLEGE OF ENGINEERING

Shavige Malleshwara Hills, Kumaraswamy Layout, Bangalore-560078
Department of Artificial Intelligence and Machine Learning



INTERNSHIP REPORT

ON

“Software Developer Intern at Aquera Labs”

Submitted in partial fulfillment for the award of degree(21INT82)

BACHELOR OF ENGINEERING IN

Department of Artificial Intelligence and Machine Learning

Submitted by:

V Jaswanth Reddy- 1DS21AI054

Conducted at

Aquera

AQUERA LABS

Department of Artificial Intelligence and Machine Learning
Dayananda Sagar College of Engineering Bangalore-78

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CERTIFICATE

This is to certify that the Internship titled “**Software Developer Intern at Aquera Labs**” carried out by **Mr V Jaswanth Reddy[1DS21AI054]**, a bonafide student of Dayananda Sagar College of Engineering, in partial fulfillment for the award of **Bachelor of Engineering**, in **Artificial Intelligence and Machine Learning** during the year 2024-2025. It is certified that all corrections/suggestions indicated have been incorporated in the report.

The internship report has been approved as it satisfies the academic requirements in respect course Internship (21INT82)

**Signature of Reporting
Head**
Ms Preethi
HR
Aquera Labs

**Signature of Internship
Coordinator**
Prof. Kavya D N
Dept. of AI & ML
DSCE

Signature of HoD
Dr. Vindhya P Malagi
Dept. of AI & ML
DSCE

External Viva:

Name of the Examiner

Signature with Date

1) _____

2) _____

DAYANANDA SAGAR COLLEGE OF ENGINEERING

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Department of Artificial Intelligence and Machine Learning



DECLARATION

I, **V Jaswanth Reddy**, final year student of Artificial Intelligence and Machine Learning, Dayananda Sagar College of Engineering, Bangalore - 560 078, declare that the Internship has been successfully completed, in **AQUERA LABS**. This report is submitted in partial fulfillment of the requirements for the award of Bachelor's degree in Artificial Intelligence and Machine Learning, during the academic year 2024- 2025.

Date:

USN: 1DS21AI054

Place: Bangalore

NAME: V Jaswanth Reddy

OFFER LETTER

Aquera

Dear Mr. Velpula Jaswanth Reddy
Mob: 8555801056

Congratulations!

We are pleased to confirm that you have been selected to work as a "Software Developer - Intern (Trainee)" for AqueraLabs India Pvt. Ltd. We are delighted to make you the following job / internship offer.

The position we are offering is that of "Intern - Backend Developer" for a period of 6 months with the stipend of Rs.25,000/- Your working hours will be from 10:00 am to 7:00 pm. However, based on the assessment of your performance during the Initial 6 months, AqueraLabs, at its sole discretion, may extend your Internship Period to a maximum of 9 months.

Only on the completion of training, meeting the Job expectations of AqueraLabs India Pvt. Ltd., you will be considered for permanent Employment with a salary package of Rs.5,60,000/- per annum based on your performance during the internship period. If your performance is not satisfactory you will be relieved from the Internship with the internship certificate for the training / internship duration.

We would like you to start training on 24 February 25. Please report to HR for documentation and orientation.

We are confident you will be able to make a significant contribution to the success of our AqueraLabs India Pvt. Ltd and look forward to working with you.

Sincerely,

For AqueraLabs India Pvt. Ltd.



Nagendra Kosur
Director of Finance & Admin

I, here by signing this as an agreement to accept the offer of AqueraLabs India Pvt. Ltd. with the terms and conditions.

Date:

Signature of the trainee:

AqueraLabs India Pvt. Ltd.
CIN No.: U72900KA2017FTC107859
Plot No. 104 P, Prestige Omega, 2nd Floor, Nallurahalli Village EPIP Zone, Bengaluru - 560066.
email: adminindia@aquera.com website: www.aquera.com

COMPLETION CERTIFICATE

ACKNOWLEDGEMENT

This Internship is a result of accumulated guidance, direction and support of several important persons. We take this opportunity to express our gratitude to all who have helped us to complete the Internship.

We express our sincere thanks to our Principal **Dr. B G Prasad**, for providing us adequate facilities to undertake this Internship.

We would like to thank **Dr. Vindhya P Malagi**, Head of Department – AI & ML, for providing us an opportunity to carry out Internship and for her valuable guidance and support.

We would like to thank **Ms Preethi**, HR, Aquera Labs , for guiding us during the period of internship.

We express our deep and profound gratitude to our Internship Co-ordinator, **Prof. Kavya D N**, Assistant Professor – AI & ML, for her keen interest and encouragement at every step in completing the Internship.

We would like to thank all the faculty members of AI & ML department for the support extended during the course of Internship.

We would like to thank the non-teaching members of the Department of AI & ML, for helping us during the Internship.

Last but not the least, we would like to thank our parents and friends without whose constant help, the completion of Internship would have not been possible.

V Jaswanth Reddy [1DS21AI054]

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CHAPTER 1

INTRODUCTION

In today's digital landscape, where organizations increasingly depend on cloud-based platforms and diverse enterprise applications, the seamless integration of identity systems has become a critical necessity. Efficient identity and access management ensures secure data exchange, automated user provisioning, and compliance with organizational policies. To achieve this, robust backend systems are required to connect various services and enable smooth interoperability.

Aquera Labs is a technology company that specializes in developing integration solutions to streamline identity management across platforms. As part of its backend development efforts, the organization focuses on building and maintaining integrations—modular components that facilitate communication between identity platforms and third-party applications. These connectors play a key role in synchronizing user data, provisioning accounts, and ensuring secure access control mechanisms.

The internship role involves supporting backend development tasks related to connector implementation. This includes contributing to code development, data transformation, and system testing. The position also requires a strong understanding of API integrations, backend architecture, and cloud-based technologies. By participating in ongoing projects, the internship provides valuable exposure to enterprise-grade software development practices, while emphasizing performance, reliability, and security.

Overall, the internship offers a practical learning environment that bridges academic knowledge with real-world application, equipping interns with relevant skills and insights into modern backend engineering within the context of identity integration.

CHAPTER 2

DESCRIPTION OF ORGANIZATION

CODTECH IT SOLUTIONS PVT LTD



Introduction

Aquera Labs is a Seattle-based technology company specializing in identity integration solutions. Founded in 2016, the company operates as a privately held entity with a global footprint, including its Indian subsidiary, **Aqueralabs India Private Limited**, headquartered in Bengaluru, Karnataka . Aquera is recognized for its innovative approach to automating identity and access management (IAM) processes across various enterprise applications.

The company's flagship offering, the **Aquera 360 HR & Identity Integration Platform-as-a-Service**, facilitates seamless synchronization of user data across HR, IT, Finance, and application ecosystems. By automating millions of Joiner, Mover, Leaver (JML) transactions daily, Aquera enhances operational efficiency, security, and compliance for over 900 clients worldwide .

Aquera's extensive library includes pre-built connectors for more than 50 HR systems and over 900 business applications, enabling rapid deployment and integration. The platform supports real-time provisioning and deprovisioning of user accounts, thereby reducing manual intervention and associated errors. Additionally, Aquera's solutions are designed to extend the capabilities of existing identity platforms, offering scalable and secure integration options for diverse organizational needs.

VISION STATEMENT

To be the leading provider of seamless identity integration solutions, enabling organizations to automate and streamline identity management processes across diverse platforms and applications.

MISSION STATEMENT

Aquera is committed to eliminating the challenges of integration around identity management platform rollouts by offering innovative solutions that simplify and accelerate the identity management provisioning and governance lifecycle. Through its advanced platform, Aquera aims to expand and expedite identity platform deployments, delivering significant time and cost savings while enhancing security and compliance for organizations worldwide.

CHAPTER 3

SCOPE OF WORK

During the course of the internship at Aquera Labs, the focus was on backend development activities related to building and maintaining identity integration connectors. These connectors enable seamless synchronization between identity management platforms and third-party applications. The role involved working closely with APIs, data transformation logic, and system automation processes to ensure reliable and secure user provisioning and deprovisioning across enterprise systems.

1. Connector Development for Identity Integration

Objective:

The objective of this project was to support the development and enhancement of connectors that facilitate identity data exchange between core identity platforms (such as Okta, Microsoft Entra ID, etc.) and external enterprise applications. These connectors ensure automated user lifecycle events such as provisioning, updates, and deactivation.

Scope:

- Assist in implementing backend logic for API-based integrations between identity systems and SaaS applications.
- Work on configuration parameters and transformation rules used to map identity attributes between systems.
- Test and validate connector performance and data flow through development and staging environments.
- Investigate integration errors and assist in debugging and resolving issues within the connector framework.

Methodology:

1. API Integration & Configuration

- Utilized REST and SCIM APIs for secure data exchange between identity platforms and applications.

- Configured connector templates with required authentication methods, endpoints, and attribute mappings.

2. Data Transformation

- Developed logic to transform identity attributes (e.g., usernames, department, roles) based on target system requirements.
- Validated transformation accuracy through simulation tools and testing environments.

3. Testing & Validation

- Executed test runs to verify data synchronization accuracy.
- Used internal logging and monitoring tools to trace and resolve issues during data exchange processes.

Results:

- Developed and tested connector logic for selected target applications.
- Ensured successful data provisioning flows with minimal error rates.
- Gained experience in connector configuration, identity attribute mapping, and API communication.

2. Automation for Connector Testing and Deployment

Objective:

The objective of this task was to support automation of routine testing procedures and enhance the deployment efficiency of connectors through scripting and backend tools.

Scope:

- Implement automated scripts to streamline connector testing workflows.
- Create validation procedures to detect data inconsistencies or mapping errors before production deployment.
- Participate in automation improvements for backend deployment pipelines.

Methodology:

1. Script Development

- Wrote reusable scripts to automate common validation tasks such as API response checking and schema validation.
- Integrated these scripts into the testing framework for pre-deployment checks.

2. Automation Integration

- Utilized internal tools and libraries to automate deployment and version control of connector updates.
- Participated in CI/CD processes and version tracking for connector releases.

3. Monitoring Enhancements

- Contributed to log configuration for better traceability of connector operations.
- Recommended improvements to error-handling routines based on testing outcomes.

Results:

- Reduced manual testing time by implementing reusable scripts.
- Improved consistency and accuracy in connector validation.
- Contributed to a more stable and efficient connector deployment process.

These projects provided hands-on experience with backend technologies, cloud-based integration, and automation practices in an enterprise context. The work done during the internship contributed to the scalability and reliability of Aquera's identity integration platform, aligning with industry best practices for secure identity data management.

CHAPTER 4

LEARNING OBJECTIVES

1. Identity Connector Development

Developing identity connectors is central to enabling secure and efficient integration between identity management platforms and third-party applications. This involves understanding protocols such as SCIM and REST, working with APIs, and implementing logic for user provisioning and deprovisioning. The process includes authentication handling, attribute mapping, and ensuring data consistency across systems. These skills are essential in enterprise environments where identity synchronization is critical for security and user experience.

2. Backend API Integration

Backend development responsibilities included integrating external APIs and building internal APIs to support connector functionality. Tasks involved structuring HTTP requests, parsing JSON responses, and ensuring secure data transactions. Emphasis was placed on creating scalable, maintainable, and testable code. API error handling, logging, and retry mechanisms were implemented to improve system reliability and fault tolerance.

3. Data Transformation and Mapping

Effective data transformation ensures compatibility between diverse systems. This includes converting identity attributes, handling complex nested structures, and applying logic for conditional transformations. Mapping data fields between identity providers and target systems required attention to detail, knowledge of data formats (JSON, XML), and transformation rules. This capability supports interoperability and smooth operation of identity workflows.

4. Cloud-Based Deployment and Testing

Experience was gained in deploying and testing connectors within a cloud-based staging environment. Tasks included unit and integration testing, using logging frameworks for debugging, and verifying data flow correctness. Testing scenarios covered edge cases, negative testing, and performance validation to ensure production readiness. These practices helped enforce software quality and compliance with enterprise-grade standards.

5. System Automation and Provisioning Logic

Automation plays a key role in maintaining connector efficiency and reliability. Learning objectives included understanding how automated provisioning and deprovisioning are triggered via event-driven mechanisms. This involved working with rule engines, sync jobs, and monitoring systems to validate operations. Automation also covered retry mechanisms and logging to ensure system transparency and auditability.

CHAPTER 5

CHALLENGES AND SOLUTION

1. Handling Missing or Inconsistent Data

Challenge:

Real-world datasets often include missing values, inconsistencies, or noise, which can compromise the integrity of analysis and reduce the accuracy of machine learning models.

Solution:

Data preprocessing techniques such as imputation, removal of null entries, and normalization were employed. This ensured a clean and consistent dataset, preserving essential patterns for effective Exploratory Data Analysis (EDA) and clustering.

2. Selecting Relevant Features for Clustering

Challenge:

Inclusion of irrelevant or improperly scaled features can result in suboptimal clustering, reducing the interpretability and reliability of segmentation.

Solution:

Feature selection was informed by insights derived from EDA and domain knowledge. Standardization techniques were applied to normalize feature scales, thereby enhancing the performance of clustering algorithms like K-Means and improving interpretability.

3. Choosing the Optimal Number of Clusters

Challenge:

Arbitrary selection of the number of clusters can lead to poor segmentation outcomes, either oversimplifying or overcomplicating the customer groups.

Solution:

The Elbow Method and Silhouette Score were utilized to evaluate and identify the optimal cluster count. These techniques provided a robust, data-driven approach for determining the most meaningful segmentation.

4. Managing API Rate Limits and Incomplete YouTube Data

Challenge:

Rate limitations and paginated responses from the YouTube Data API posed difficulties in retrieving a complete set of video comments efficiently.

Solution:

Efficient pagination handling and retry logic were implemented. This ensured comprehensive data collection within API constraints, enabling reliable sentiment analysis of YouTube comments.

5. Ensuring Sentiment Classification Accuracy

Challenge:

Generic sentiment analysis models may not accurately interpret context-specific language, slang, or tone typical of YouTube comment sections.

Solution:

A transformer-based model from Hugging Face was employed to deliver high-accuracy sentiment classification. Threshold adjustments were made to better distinguish neutral sentiments, enhancing classification precision and insight quality.

6. Preventing Duplicate Storage in Firestore

Challenge:

Storing duplicate comments in Firestore can lead to inefficient use of storage, data redundancy, and inaccurate analysis.

Solution:

A validation step was introduced to check for existing comment IDs prior to insertion. This ensured that only unique comments were stored, preserving data integrity and optimizing database performance.

7. Automating Tasks Without Interruptions

Challenge:

Long-running automated scripts are vulnerable to interruptions, memory leaks, and API failures, impacting the continuity of real-time sentiment analysis.

Solution:

The automation pipeline incorporated a robust looping mechanism with controlled delay intervals, exception handling, and error logging. This design allowed the system to self-recover from failures and maintain continuous operation with minimal oversight.

HARDWARE AND SOFTWARE REQUIREMENTS

Software requirements:

- **Operating System:** Windows 10 (64-bit)
- **Programming Language:** Java Script(For scripting, data preprocessing, and API handling)
- **Libraries & Frameworks:**
 - **Node 20** (For backend run time environment)
 - **Jest** (For unit testing)
- **Development Tools:**
 - **Postman** – for testing out API calls
 - **Visual Studio Code / Terminal** – for coding and running automation scripts

Hardware Requirements:

- Processor:** Intel Core i5 / AMD Ryzen 5 or higher(Sufficient for running multi-threaded tasks and development environments)
- RAM:** Minimum 8 GB(To ensure smooth performance during data processing and model inference)
- GPU(Recommended):** NVIDIA RTX 3050 / A100 / V100(*Useful for faster execution*)
- Storage:** At least 256 GB SSD(For code repositories, libraries, datasets, and virtual environments)
- Internet:** High-speed fiber optic internet(Required for accessing cloud APIs, real-time data from YouTube, and Firebase services)
- **Connectivity:** LAN Port (for secure and stable internet access)
- **Input Devices:** Standard Mouse and Keyboard

CHAPTER 6

ACHIEVEMENTS AND CONTRIBUTION

1. Backend Integration

- **End-to-End Development Exposure:** Contributed to backend development tasks involving API integrations, data handling, and connector workflows. This work supported seamless communication between identity systems and external applications.
- **Implementation of Backend Logic:** Assisted in implementing and debugging logic for various operations such as user provisioning, updates, and deletions, ensuring accuracy and consistency across data transactions.
- **Integration with External APIs:** Worked on consuming and processing data from multiple APIs, handling request formatting, response parsing, and error management to support scalable connector functionality.
- **Configuration-Based Architecture Understanding:** Gained exposure to configurable connector frameworks, understanding how business rules and field mappings drive connector behavior in dynamic enterprise environments.
- **System Design Awareness:** Understood modular backend designs that allow individual components like sync logic, transformations, and logging to be developed and maintained independently.
- **Automation and Monitoring:** Contributed to scheduled and event-driven automation processes that streamline data synchronization between systems with minimal manual intervention.
- **Technical Problem Solving:** Participated in identifying and resolving common backend issues such as data mismatches, transformation errors, and integration bugs.
- **Documentation and Best Practices:** Helped in preparing internal notes, use-case summaries, and configuration outlines to support knowledge transfer and future maintenance.

- **Schema Mapping and Attribute Transformation:** Assisted in defining and refining attribute mappings between source and target systems, ensuring that identity data was correctly transformed and transmitted during provisioning processes.
- **Version Control and Collaboration:** Worked with version control systems like Git to manage code changes and collaborate effectively with the development team, following standard branching and code review practices.

CHAPTER 7

REFLECTION AND ANALYSIS

This experience offered a thorough exploration of the critical role customer satisfaction plays in business success, emphasizing both interpersonal skills and process improvements that drive positive customer experiences. Engaging directly with customers highlighted the multifaceted nature of satisfaction, which depends not only on product quality but also on communication, responsiveness, and problem resolution. The project deepened my understanding of how customer feedback loops and service excellence contribute to sustained loyalty and business growth.

One of the most valuable lessons was the significance of proactive communication and timely support. By addressing customer inquiries swiftly and accurately, I helped minimize frustration and build trust. This proactive approach was crucial in preventing minor issues from escalating into significant problems, underscoring the importance of responsiveness in customer relations.

Focusing on personalized service allowed me to tailor solutions to specific customer needs, enhancing the overall experience. This required active listening and empathy, skills that proved essential in understanding customer pain points and delivering relevant assistance. Implementing clear documentation and user guides further empowered customers, reducing their dependency on support and increasing satisfaction through self-service options.

Moreover, integrating customer feedback into continuous improvement cycles was instrumental. Analyzing feedback trends enabled me to identify recurring issues and propose actionable enhancements, thereby improving product quality and service processes. This feedback-driven mindset demonstrated the value of listening to customers as active partners in shaping the service experience.

Building long-term relationships was another important contribution. Consistent follow-ups and genuine engagement helped nurture loyalty and encouraged repeat business. The emphasis on relationship management highlighted how trust and rapport directly influence customer retention and positive word-of-mouth referrals.

Overall, this experience reinforced the interconnected nature of customer satisfaction components—communication, quality, responsiveness, and feedback integration—and how my contributions in each area collectively elevated the customer experience. This comprehensive approach to customer

satisfaction is essential for creating lasting value and will inform my future efforts to foster strong customer-centric cultures in professional environments.

CHAPTER 8

RECOMMENDATIONS

- **Enhance Customer Feedback Channels:** Develop more diverse and accessible feedback mechanisms such as in-app surveys, chatbots, and social media monitoring to capture a broader range of customer insights. This will enable a more comprehensive understanding of customer needs and expectations.
- **Implement Proactive Support Systems:** Utilize automated alerts and AI-driven support tools to identify potential issues before customers report them. Proactive engagement can significantly reduce resolution time and improve overall satisfaction.
- **Personalize Customer Interactions:** Leverage CRM data and analytics to tailor communications and offers based on individual customer preferences and behaviors. Personalization fosters stronger relationships and increases customer loyalty.
- **Streamline Issue Resolution Processes:** Establish clear escalation paths and empower frontline staff with decision-making authority to resolve problems swiftly. Efficient problem-solving enhances trust and reduces churn.
- **Regular Training for Support Teams:** Invest in ongoing training focused on empathy, communication skills, and product knowledge. Well-prepared teams provide higher quality service, directly impacting customer satisfaction.
- **Use Data Analytics to Identify Pain Points:** Analyze customer service data to spot recurring issues and bottlenecks. Applying these insights to refine processes and products leads to continuous improvement.
- **Follow-up and Relationship Management:** Implement systematic follow-ups after service interactions to ensure customer satisfaction and gather additional feedback. This reinforces commitment to customer care and encourages repeat business.

- **Foster a Customer-Centric Culture:** Promote company-wide awareness of customer satisfaction metrics and success stories. Encouraging collaboration across departments ensures aligned efforts to improve the customer experience.

CHAPTER 9

CONCLUSION

This internship provided a comprehensive exposure to the practical application of data analysis, machine learning, and cloud computing within a professional setting. Beginning with foundational tasks such as data exploration and preprocessing, I was able to uncover meaningful insights that guided the development of analytical models and solutions tailored to business needs. This hands-on experience emphasized the value of structured data workflows and informed decision-making.

Throughout the internship, I worked on implementing machine learning techniques to address real-world challenges, gaining experience in model selection, evaluation, and optimization. The application of natural language processing techniques further enriched my understanding of handling unstructured data, enabling deeper insights into user behavior and sentiment. Leveraging modern tools and APIs facilitated dynamic data collection and continuous analysis, mirroring industry best practices.

By integrating cloud-based storage and automation, I observed the critical role of scalable and reliable systems in managing data pipelines for ongoing operations. Automating routine processes not only improved efficiency but also ensured consistent data handling with minimal manual intervention. These experiences highlighted the importance of combining technical knowledge with software engineering principles to deliver robust solutions.

Beyond technical skills, the internship honed my problem-solving abilities and adaptability in a fast-paced environment. Navigating challenges such as data quality issues, system constraints, and evolving project requirements strengthened my capacity for iterative learning and solution refinement. This comprehensive experience has laid a strong foundation for future projects involving data-driven decision-making and technology integration.

Overall, the internship was a well-rounded journey that enhanced both my technical expertise and professional growth. It showcased the importance of interdisciplinary approaches in solving complex problems and underscored the potential of data science and cloud technologies to drive meaningful business impact.

CHAPTER 10

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